

Interactive 3D-Printed Soap: Exploring Biodegradable and Ephemeral Material Interactions

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Figure 1: 3D printing soap and interactive artifacts. (a) Custom DIY 3D printer with a gravity-fed extruder for soap 3D printing. (b) Color-changing vase printed with thermochromic and photochromic powders. (c) Temporal QR code in scented soap that blurs and dissolves with use. (d) pH-sensing ring using turmeric for visual water-quality feedback. (e) Ephemeral timer circuit with activated carbon soap: LED brightness varies with wetting and disappears as the soap dissolves.

Abstract

Soap is a colorful, DIY-friendly, and biodegradable material with potential beyond cleaning. This demonstration presents a diverse design space of 3D-printed soap, functionalized with additives for customized color, scent, conductivity, and environmental responsiveness. Fabricated with a low-cost printer with a custom extruder optimized for soap, these artifacts

are presented for direct interaction on-site. Participants can explore their expressive and functional possibilities, experiencing soap's temporal materiality and discovering its potential as a sustainable, environmentally responsive medium for creative and playful interaction design.

Keywords

Sustainable 3D printing, digital fabrication, biodegradable material, sustainable HCI

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1 INTRODUCTION AND BACKGROUND

Traditionally, soap is produced through saponification [1, 6], a chemical process combining oils, lye, and water. While it solidifies at room temperature and dissolves easily in water, shaping soap has relied mainly on casting methods such as cold processing or melt-pour molds. Despite its long history in both industrial and DIY making practices [8], soap has received little attention as a 3D-printable material in Human-Computer Interaction (HCI) and digital fabrication research. Yet, its combination of everyday accessibility and water-solubility as a biodegradable cleaning agent, combined with its rich design potential, including variations in color, form, scent, and functional properties (e.g., therapeutic or exfoliating effects), makes soap a compelling medium for creative and interactive fabrication. Motivated by growing interest in sustainable and ephemeral materials, this work explores 3D-printed soap as a medium for material-driven interactive fabrication.

Our work is motivated by the growing interest in biomaterials for digital fabrication. Prior studies have demonstrated how substances like edible pastes [11, 21], starch-based dough [3], and recycled waste such as seashell [19] or eggshell paste [2] can introduce new textures and material experiences. Beyond structural use, researchers have leveraged the degradability of gelatin [13, 20], sodium alginate [14], and PVA [5, 16] to integrate water-solubility into interaction design for removal or reconfiguration. Extending this line of inquiry, we propose soap as a novel and sustainable for 3D-printing material whose biodegradability and reusability support sustainable making while opening opportunities for interaction design leveraging its transient properties.

We repurpose solid bar soaps into printable formulations, enabling customization of color, scent, and environmental responsiveness (e.g., pH, UV, temperature). Soap printed artifacts harden naturally by air-drying, requiring no special post-processing. Unlike conventional 3D printing materials, soap's ephemeral quality allows gradual dissolution, creating tangible, time-based interactions. Our demonstration presents a hands-on exploration of 3D Printing soap's design space. Participants can interact with printed artifacts that reveal information, color, or transform over time, experiencing material change as a direct component of engagement. This work highlights soap as a playful, sustainable, and expressive medium for ephemeral and environmentally responsive interaction design.

2 MATERIAL SYSTEM AND FABRICATION OVERVIEW

To enable 3D printing of soap, we developed a formulation specifically designed for extrusion and structural stability. The material combines a soap base—either 100g of repurposed bar soap or a custom-prepared mixture of 100g plant oils,

14g sodium hydroxide, and 33g water—with 40g calcium carbonate, 3g sodium alginate, and 80g water. Functional additives—including thermochromic [10] and photochromic powders [9], natural pigments such as turmeric [4, 12], and activated carbon [18]—were incorporated at 5% by weight to provide color variation, environmental responsiveness, pH sensing, and electrical conductivity. Using standard cold-process methods [15], all sodium hydroxide is fully consumed during saponification, resulting in a skin-safe, non-toxic, and fully biodegradable material. Printed artifacts can be reused or safely dissolved in water, supporting sustainable and iterative experimentation.

The 3D printing soap fabrication workflow which is shown in (Figure 2) and is conducted at room temperature using a custom, low-cost gravity-fed extrusion printer with a plunger-driven 2mm nozzle mounted on a Cartesian gantry system (Figure 4, a-2,4,5) built by adapting open-source designs [7]. Print parameters are calibrated to the paste's rheology to ensure consistent deposition, layer adhesion, and dimensional fidelity. Digital models are converted into precise toolpaths using OrcaSlicer, an open-source slicing program [17], matched to the material's flow characteristics to produce reliable and reproducible prints. This integrated material-hardware system establishes a robust platform for exploring ephemeral, interactive soap-based artifacts, enabling functional augmentation, iterative design exploration, and environmentally responsible practices. We explored ways of customizing soap-based materials by adjusting their color and fragrance, and further expanded their capabilities by introducing electrical conductivity (Figure 3). To widen the design space, we integrated responsive additives that react to environmental stimuli such as temperature, UV exposure, and pH changes. These properties enable new forms of interaction for 3D-printed soap artifacts. Throughout the process, all materials were selected to remain biodegradable and non-toxic.

3 INTERACTIVE ARTIFACTS AND EXPERIENCES

To explore the expressive and functional potential of 3D-printed soap, we created a range of interactive artifacts leveraging its multifunctional properties—color variation, conductivity, environmental responsiveness, and ephemeral dissolution—within everyday contexts such as handwashing, household objects, and wearable forms. Our demonstration invites participants to directly interact with these artifacts, observing, manipulating, and experiencing how the material changes in real time, making transformation a core part of the engagement.

Participants can engage directly with these artifacts, experiencing how soap's temporal materiality shapes interaction and perceiving the interplay between material, function, and environment. For example, a temporal QR code (Figure 1-c) grants participants short-lived access to digital resources: as the soap is used, the code gradually blurs and dissolves, naturally revoking access while providing a tangible sense of temporality. An ephemeral timer circuit (Figure 1-e) further



Figure 2: Workflow for 3D printing soap artifacts. (a) Objects are modeled in Rhino, (b) sliced in OrcaSlicer with a profile tuned for soap pastes, and (c) converted into G-code and set up the printer, (d) the printer's gravity-fed hopper is filled with the prepared formulation. (e) The system deposits the soap paste layer by layer to build the object, then (f) the printed artifact is left to air-dry at room temperature for 12 hours, (g) the resulting biodegradable artifacts exemplify ephemeral material-based fabrication, integrating digital design, material preparation, and custom hardware.

Additive	Recipe	Sample	Sample Printed
	Soap base+Printing Enhancer: 40g Calcium carbonate 0g glycerin 3g Sodium alginate 80g water 2 drop green food coloring		
	Soap base+Printing Enhancer: 40g Calcium carbonate 0g glycerin 3g Sodium alginate 80g water 5% thermochromic (by weight)		
	Soap base+Printing Enhancer: 40g Calcium carbonate 0g glycerin 3g Sodium alginate 80g water 5% turmeric (by weight)		
	Soap base+Printing Enhancer: 40g Calcium carbonate 0g glycerin 3g Sodium alginate 80g water 5% UV photochromic (by weight)		
	Soap base+Printing Enhancer: 40g Calcium carbonate 0g glycerin 3g Sodium alginate 80g water 5% activated carbon (by weight)		

Figure 3: Customizable functional soap recipes. Each formulation combines a soap base (calcium carbonate, glycerin, sodium alginate, water) with additives that provide specific functionalities: (from top) color, thermal sensing, pH sensing, UV sensing, and conductivity.

externalizes time, as water-activated conductivity increases the LED brightness until the dissolving soap eventually breaks the circuit, transforming a routine meditation ritual into a participatory and time-bound experience. Environmental responsiveness is showcased in a color-changing vase (Figure 1-a), where thermochromic and photochromic powders allow participants to witness dynamic color shifts in response to temperature and UV exposure, while a pH-sensing wearable ring (Figure 1-d) provides immediate visual feedback on water acidity during handwashing. Across these examples, participants experience multisensory interactions—combining sight,

touch, and scent—while exploring soap's potential for reuse, remolding, and creative repurposing. Collectively, these artifacts illustrate how ephemeral, biodegradable materials can mediate interactions in which material change itself becomes a core component of engagement, providing a rich platform for playful, reflective, and environmentally responsive experiences in a hands-on demo setting.

4 CONCLUSION

This demonstration highlights 3D-printed soap as a sustainable, ephemeral medium for interactive artifact design. By

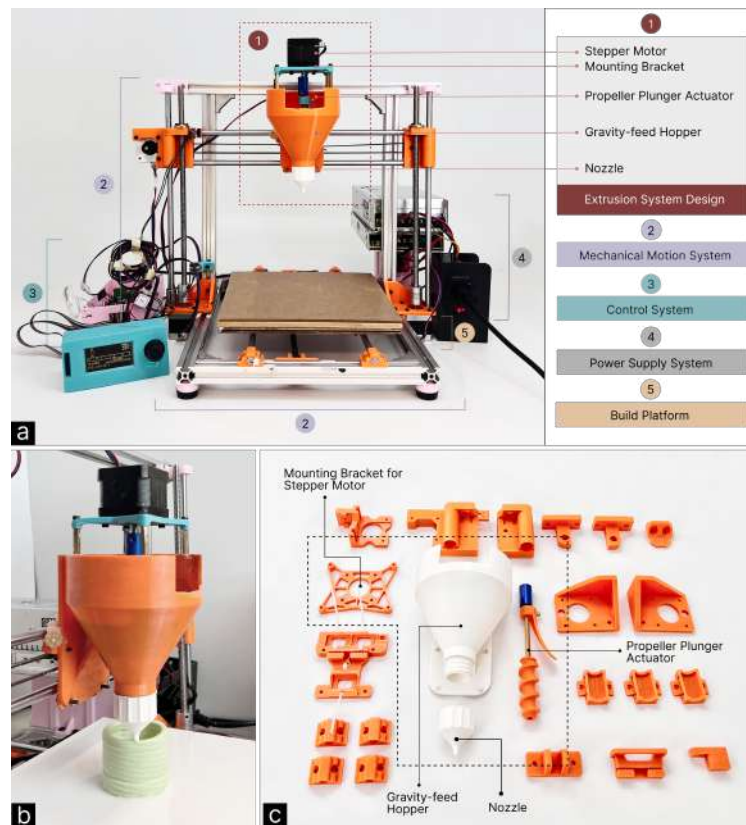


Figure 4: (a) Labeled view of the custom-built soap 3D printer: The system includes a Cartesian gantry, extrusion system, control system, power supply and build platform, using open-source mechanical components and Marlin firmware to build. (b) Close-up of the extrusion system design in printing, showcasing the gravity-fed hopper, rotating plunger actuator, and 2mm nozzle. (c) Disassembled view of the printer for soap based material, highlighting key components of the extrusion system.

co-developing soap formulations with a custom low-cost extrusion printer, we establish a material–hardware system that supports expressive, environmentally responsive interactions, including color change, pH sensing, conductivity, and temporality. The artifacts show how ephemeral materials can transform everyday routines into reflective, participatory experiences, providing designers with a versatile platform to explore temporally and materially mediated interactions. By situating biodegradable materials within digital fabrication, this work expands the design space for playful, interactive, and contextually situated experiences, foregrounding sustainability, sensory engagement, and the creative potential of ephemeral materials.

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